國立臺灣師範大學課程綱要【預測學習】
National Taiwan Normal University Syllabus

一、 課程基本資料（授課教師不可異動部分，同一課程名稱此部分應相同）

I. Course information (maintained by Information Technology Center)

科目代碼 **Course Code**		**課程名稱** **Chinese Course Name**	預測學習	
英文課程名稱 **Course Name**	Predictive Learning From Data			
全/半年 **Two/One semester**	半年 (One semester)		**必修/選修/通識** **Required/Elective/ General Education**	選修 (Elective)
總學分數 **Credits**	3		**每週上課時數** **Weekly Lecturing hours**	3
開課系所 **Department/ Program**	資工系/大碩 (EMI)	**先修科目** **Prerequisite Course**		
授課教師 **Instructor(s)**	陳翔瀚 (Hsiang-Han Chen)			
課程簡介 **（中/英文）** **Course Description (Both of English and Chinese are required.)**	（英文） This course presents predictive data-analytic modeling methods, which aim to estimate a model providing generalization, i.e., good prediction for unseen data. These methods have been developed in various disciplines, e.g., statistics, machine learning, pattern recognition, data mining, etc. Three levels of predictive data-analytic modeling methods, including conceptual/mathematical, technical, and philosophical, would be discussed. （中文）			

	本課程介紹預測性數據分析建模方法，其目標為建立具良好泛化性之預測模型(亦即對未知數據具有良好預測能力)。此類方法已在多個學科領域得到良好的發展，包括統計學、機器學習、模式識別、數據挖掘等。本課程將分三個層次討論預測性數據分析建模方法，包括數學概念、技術和哲學。

課程目標 **Curriculum Goals**		**對應系所核心能力** **Corresponding to the Departmental Core Goal**
1	針對資料/問題選擇適合的預測模型與建模方法 (for different problems/data, the student can select suitable predictive models and modeling methods)	1.3 能熟悉資訊基礎理論 (Acquainted with foundations of computer science and mathematics) 2.3 具有從經驗中提升專業思考層次的能 力 (Capability to enhance professional thinking from experience)
2	針對真實世界問題建置具良好泛化性之預測模型 (the student can build predictive models with good generalization to real world problems)	2.1 具有資訊系統發展分析與整合的能力 (Capability to develop and integrate computer systems) 2.2 具有計畫執行與團隊合作的能力 (Capability to execute projects and skill set of teamwork)
3	能在課堂上進行報告與討論 (the student can give presentations and organize discussions in class)	3.2 能有良好的溝通與表達能力 (Good communication and expression skill set)

二、 教學大綱（授課教師可異動部分）

II. General Syllabus (maintained by instructors each semester)

授課教師 **Instructor(s)**	陳翔瀚 (Hsiang-Han Chen)

1. **教學進度與主題**

Course Schedule and Topic

Week #	Topic
1	Introduction and overview
2	Formulation of the learning problem & classical methods
3	Formulation of the learning problem & classical methods
4	Regularization and complexity control
5	Statistical learning theory
6	Statistical learning theory
7	Nonlinear optimization strategies
8	Midterm
9	Methods for data reduction and dimensionality reduction
10	Methods for data reduction and dimensionality reduction
11	Methods for regression
12	Methods for classification
13	Margin based methods and support vector machines
14	Advanced / non-standard learning settings
15	Final Presentation
16	Final Presentation

2. 教學方法

Teaching Approach

方式 **Methods**	說明 **Notes**
■ 講述法 **Formal Lecture**	Formal Lectures help students to understand the course materials.
□ 討論法 **Group Discussion**	
□ 問題解決教學 **Problem-based Learning**	
■ 合作學習 **Cooperative Learning**	Cooperative learning by encouraging students to work together to solve research problems in course projects.
■ 實驗/實作	Practice by encouraging students to implement the knowledge learned from the course in their course projects.

Practice	
☐ **實地考察、參訪** **Fieldwork**	
☐ **媒體融入教學** **Multimedia Lecture**	
☐ **專題研究** **Case Studies**	
☐ **其他** **Other**	

3. 評量方法

Evaluation Criteria

方法 **Methods**	百分比 **Percentage**	說明 **Notes**
■ **作業** **Assignments**	30%	Homework assignments
■ **期中考** **Midterm Exam**	25%	Course project - proposal presentation
■ **期末考** **Final Exam**	35%	Course project - final presentation (20%) + written report (15%)
☐ **課堂討論參與** **Class Discussion and Participation**		
■ **出席** **Attendance**	10%	Participation

☐ 報告 **Presentation**		
☐ 成果展覽 **Shows/Exhibitions**		
☐ 專題 **Case Study Reports**		
☐ 其他 **Other**		
4. 參考書目 **Course Material and Reference**	1. V. Cherkassky and F. Mulier, Learning from Data. Concepts, Theory and Methods, Second edition, Wiley-Interscience 2007 2. V. Cherkassky, Predictive Learning, 2013 http://vctextbook.com/	